

Hepatitis-C-Classification-and-Clustering

Project Overview:

Hepatitis C is a viral infection that causes liver inflammation, sometimes leading to serious liver damage. The hepatitis C virus (HCV) spreads through contaminated blood.

The infection was one of the main death causes from 1960 to 1970. In this project, we will be studying a dataset of Hepatitis C infection. The dataset was donated by the Medical University Hannover (MHH); Hannover, Germany.

You will walk through an example containing some data preparation, classification task and a clustering task to end up with a conclusion part where you will be comparing the two methods and providing explanation about the observed phenomenon.

Details:

In this project, you will be :

- Loading the data and cleaning it to be ready for the machine learning tasks (cleaning includes : data types checks, missing values....)
- You will be clustering the data without taking into account the provided target, then storing those clusters as a new feature in the dataset.
- You will be building a model to classify this data, evaluate it on a test set then storing the new test set, so that it includes : the clusters, the targets and the predicted targets.
- Finally, you will compare the obtained clusters' composition with both the actual targets and the predicted ones using the previously stored test set. You have to provide an explanation about your analysis.